SELLERS TABLE

```
sellers=pd.read_csv(os.path.join('/kaggle/input/datasets/organizations/olistbr/brazilian-ecommerce/ol
sellers.head()
```

	seller_id	seller_zip_code_prefix	seller_city	seller_state
0	3442f8959a84dea7ee197c632cb2df15	13023	campinas	SP
1	d1b65fc7debc3361ea86b5f14c68d2e2	13844	mogi guacu	SP
2	ce3ad9de960102d0677a81f5d0bb7b2d	20031	rio de janeiro	RJ
3	c0f3eea2e14555b6faeea3dd58c1b1c3	4195	sao paulo	SP
4	51a04a8a6bdcb23deccc82b0b80742cf	12914	braganca paulista	SP

SELLERS ISNULL

```
sellers.isnull().sum()
```

```
seller_id                 0
seller_zip_code_prefix    0
seller_city               0
seller_state              0
dtype: int64
```

CUSTOMERS ISNULL

```
customers.isnull().sum()
```

```
customer_id                 0
customer_unique_id          0
customer_zip_code_prefix    0
customer_city               0
customer_state              0
dtype: int64
```

CUSTOMERS DUPLICATES

```
customers.duplicated().sum()
```

```
np.int64(0)
```

CUSTOMERS TABLE

	customer_id	customer_unique_id	customer_zip_code_prefix	customer_city	customer_state
0	06b8999e2fba1a1fbc88172c00ba8bc7	861eff4711a542e4b93843c6dd7febb0	14409	franca	SP
1	18955e83d337fd6b2def6b18a428ac77	290c77bc529b7ac935b93aa66c333dc3	9790	sao bernardo do campo	SP
2	4e7b3e00288586ebd08712fdd0374a03	060e732b5b29e8181a18229c7b0b2b5e	1151	sao paulo	SP
3	b2b6027bc5c5109e529d4dc6358b12c3	259dac757896d24d7702b9acbbff3f3c	8775	mogi das cruzes	SP
4	4f2d8ab171c80ec8364f7c12e35b23ad	345ecd01c38d18a9036ed96c73b8d066	13056	campinas	SP

CUSTOMER CITY

```
customers['customer_city_clean']=customers['customer_city'].apply(
    lambda x: ''.join(char for char in str(x) if ord(char)<128)
)
customers[['customer_city','customer_city_clean']].head()
```

	customer_city	customer_city_clean
0	franca	franca
1	sao bernardo do campo	sao bernardo do campo
2	sao paulo	sao paulo
3	mogi das cruzes	mogi das cruzes
4	campinas	campinas

EXECUTIVE SUMMARY

```
print("=== EXECUTIVE SUMMARY ===")
print("Total Orders:", orders['order_id'].nunique())
print("Total Revenue:", round(order_items['price'].sum(),2))
print("Average Delivery Time:", round(orders['delivery_time_days'].mean(),2))
print("Late Delivery %:", round(orders['is_late'].mean()*100,2))
```

```
=== EXECUTIVE SUMMARY ===
Total Orders: 99441
Total Revenue: 13591643.7
Average Delivery Time: 12.09
Late Delivery %: 7.87
```

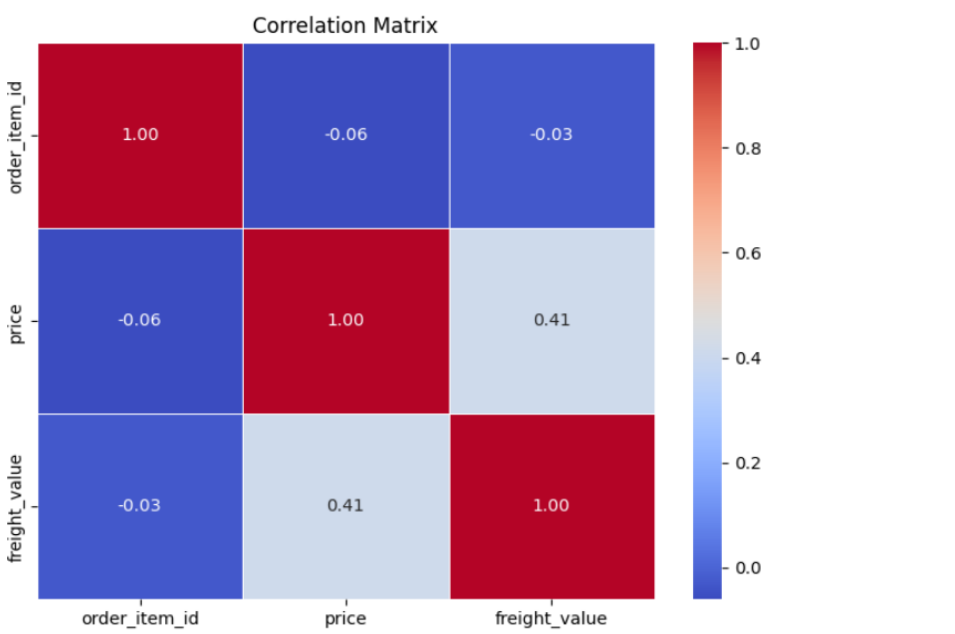
SORT VALUES BY CITIES

```
#sort cities based on ascii
customers_sorted=customers.sort_values(by='city_first_letter_ascii')
customers_sorted.head()
```

	customer_id	customer_unique_id	customer_zip_code_prefix	customer_city	customer_state	customer_city_clean	city
99351	f83083fcbc51d12f8279b5713c2d4b4d	0465f164e35aba70f70e911b252513a9	77807	araguaina	TO	araguaina	
99406	d9110683c7a282144e9fc97660026a28	5cbfdb85ec130898108b32c50d619c39	74980	aparecida de goiania	GO	aparecida de goiania	
73180	d2da057d4083d5150c6275d1d17b7070	3dadc7103d20bae9a1d9e0ade0314b5e	28970	araruama	RJ	araruama	
73244	891966aa607cc04bf5575d6b22ef9e57	d1da2c3c54238beabf857d2d7fe4e8ab	29230	anchieta	ES	anchieta	
73239	2f7001da9e597c4b707313a341a20614	38c576114024f3c3e45f83e7b5ee0c56	59965	alexandria	RN	alexandria	

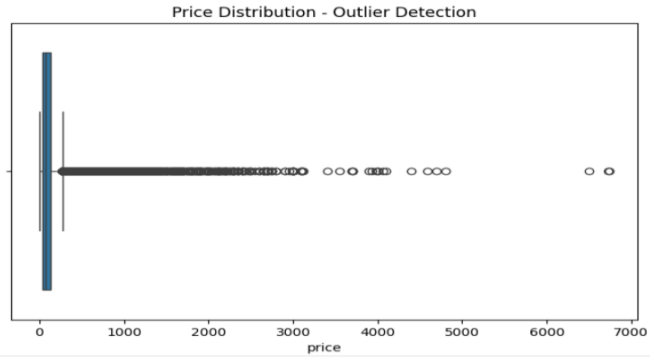
CORRELATION MATRIX

```
plt.figure(figsize=(8,6))
sns.heatmap(numeric_cols.corr(),annot=True, cmap='coolwarm',fmt=".2f",linewidths=0.5)
plt.title("Correlation Matrix")
plt.show()
```



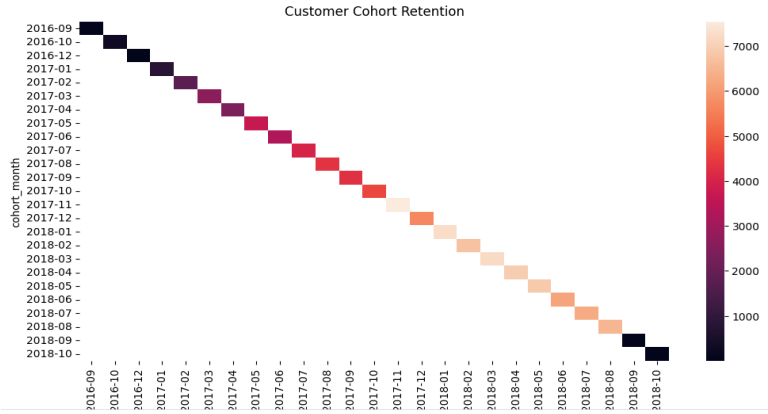
DISTRIBUTION PLOT

```
plt.figure(figsize=(8,5))
sns.boxplot(x=order_items['price'])
plt.title("Price Distribution - Outlier Detection")
plt.show()
```



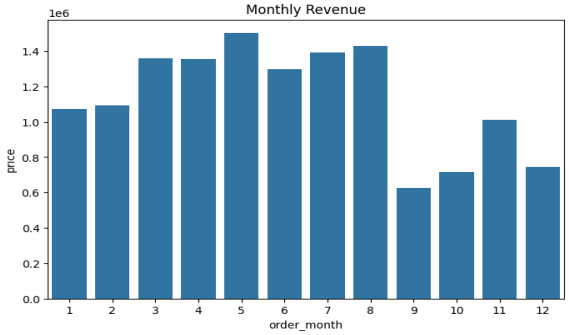
HEATMAP COHORT RETENTION – CUTOMERS

```
plt.figure(figsize=(12,6))
sns.heatmap(cohort_pivot,annot=False)
plt.title("Customer Cohort Retention")
plt.show()
```



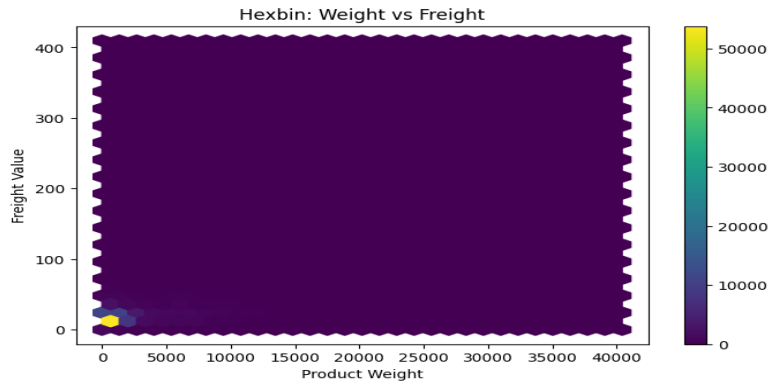
MONTHLY REVENUE

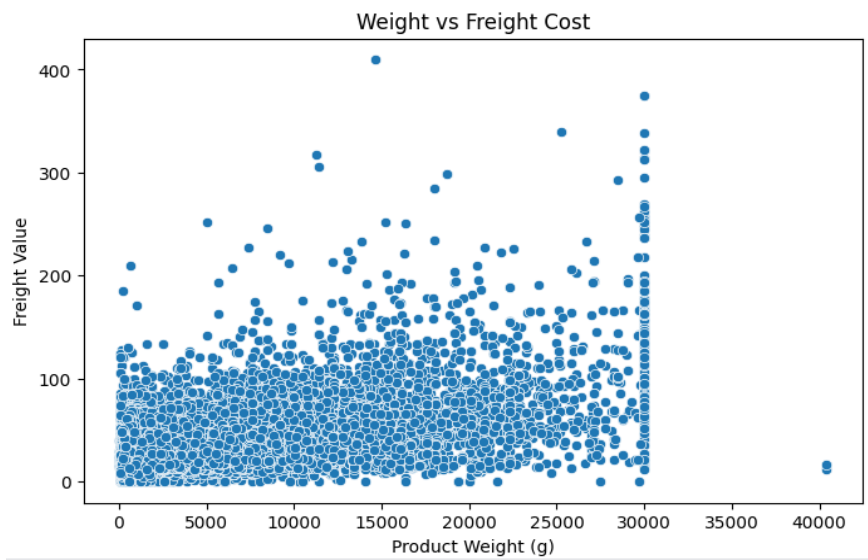
```
monthly_revenue=orders_revenue.groupby('order_month')['price'].sum().reset_index()

plt.figure(figsize=(8,5))
sns.barplot(data=monthly_revenue, x='order_month',y='price')
plt.title("Monthly Revenue")
plt.show()
```



HEXBIN

```
plt.figure(figsize=(8,5))
plt.hexbin(product_merge['product_weight_g'],product_merge['freight_value'],gridsize=30)

plt.xlabel("Product Weight")
plt.ylabel("Freight Value")
plt.title("Hexbin: Weight vs Freight")
plt.colorbar()
plt.show()
```



SCATTERED PLOT

```
plt.figure(figsize=(8,5))
sns.scatterplot(data=product_merge, x='product_weight_g',y='freight_value')

plt.title("Weight vs Freight Cost")
plt.xlabel("Product Weight (g)")
plt.ylabel("Freight Value")
plt.show()
```